



4810 - Quantitative Study on the Search for Sulfur-bearing Molecular Ice

Signatures

Cycle: 3, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

JWST programs #1309 and #2151 published new and pioneering findings on the composition and abundance of ices in the Chameleon I interstellar molecular cloud and protostellar environment (Yang et al. 2022; McClure et al. 2023). Notably, JWST observations confirm that the sulfur (S)

budget – which S is one of the 6 biogenic elements – is tremendously low in the cloud (1% of the cosmic abundance), with OCS as the only S-bearing molecule securely detected towards the cloud and protostar. With other S-bearing molecules, such as H_2S and SO_2 detected in the gas-phase in clouds and around protostars, in addition to other justifications, it is suspected that S-bearing ice signatures are hidden, rather than missing, in the archived data.

We propose to perform laboratory experiments, supported by computational simulations, of iron sulfide-ice reactions for a resilient search for H_2S and SO_2 ice signatures in the Chameleon I cloud and protostellar data of #1309, #2151 and #1854. Sulfide minerals are proposed to be significant sources of sulfur in the Chameleon I cloud (McClure et al. 2023). Deliverables include quantitative results, such as column densities and abundances of reactants, intermediates and products to compare to the archived data. This quantitative data will be critical to the interpretation of data from other JWST programs, such as JOYS (#1290), which preliminary spectral studies towards >30 protostars have indicated the necessity for the work proposed here (private communication).

OBSERVING DESCRIPTION

N/A